ABAP INTERVIEW PREPARATION

ABAP practice workbook

Thirteen interview scenarios with model answers, a mock interview and a study plan

329 questions across 24 study segments. Reviewed 11 October 2026.

Use the short answers for oral rehearsal. The examples and pitfalls prepare you for follow-up questions. Choose the target SAP edition and release before applying an implementation detail.

Read online in GitHub through README.md, or download the repository and open START_HERE.html for offline recall cards. The PDF is a portable reading and printing edition.

Solve each exercise before opening the suggested answer. The scenarios are hypothetical interview exercises. Code fragments illustrate reasoning and are not activated SAP repository objects. Validate exact syntax, dependencies and release compatibility in a target SAP system before using a fragment in an application.

Exercise 1 Choose an internal table

You receive 100,000 unique order IDs. For each of 40,000 incoming messages, you must find the order by its complete ID. In another report you must process all orders for each customer in customer order. Choose a table category and explain the trade-offs.

Your answer before reading the solution

Model answer

For repeated complete unique-key lookups, consider a hashed table with the order ID as unique key. For grouped ordered processing, use a sorted table with a suitable leading customer key or a standard table with a named sorted secondary key. Account for writes, memory, key-maintenance cost and whether input order must be preserved. Do not describe partial hashed-key access as constant time.

Follow-up

What changes when duplicate order IDs are permitted or only a partial key is known? Explain how you would handle a missing row without hiding a business error. Review ABAP-01 through ABAP-04.

Exercise 2 Find the data-access defect

An import selects details with FOR ALL ENTRIES using an internal table of selected document IDs. The selection table can be empty. The developer also assumes duplicate result rows will be preserved.

Your answer before reading the solution

Model answer

Guard against an empty driver before issuing the query; otherwise the FOR ALL ENTRIES condition can be ignored and a much larger result retrieved. Consider whether a join, range or another supported query better expresses the requirement. FOR ALL ENTRIES also removes duplicate result rows, so the projected columns and intended multiplicity matter. Deduplicate a driver only when that preserves the intended semantics and reduces redundant work.

Follow-up

Do not promise that a join is always faster. Compare result equivalence, transferred rows, execution count and representative timings. Review the SQL segment.

Exercise 3 Correct a CDS identity

A flight entity exposes CARRID and CONNID as its only keys while reading SFLIGHT. It associates to SCARR using the carrier ID, declares a to-many relationship and projects `_Carrier.name`. The client occasionally confuses different departure dates.

Your answer before reading the solution

Model answer

Include FLDATE in the identity because departures can share carrier and connection but differ by date. Verify the actual SCARR element, CARRNAME, rather than a guessed `name` element. Model the carrier relationship with truthful to-one cardinality according to the ON condition and completeness of data. Do not claim a CDS key enforces database uniqueness. For a new supported model prefer a view entity; omit the legacy SQL-view-name annotation.

Follow-up

Test multiple dates, a missing carrier and unexpected duplicate target rows. Decide DCL at the actual consumed entity. Review CDS-02, CDS-03, CDS-07 and CDS-11. This replaces the malformed illustrative CDS snippet in the original PDF rather than offering it as a ready-to-activate object.

Exercise 4 Repair RAP transaction ownership

A proposed action changes a request, attempts COMMIT ENTITIES, then would invoke a remote workflow. That nested commit violates the RAP runtime contract. The design also needs to reject invalid date ranges before persistence. The developer proposes moving the commit into the saver.

Your answer before reading the solution

Model answer

Remove the independent commit from the behavior implementation. The RAP framework owns transaction completion; moving it to a saver does not fix the violation. Derive data through appropriate behavior, reject invalid instances before persistence, and coordinate workflow publication with committed state through a supported transactional mechanism. The external EML consumer, if present, checks modification responses and owns the outer COMMIT ENTITIES.

Follow-up

Test invalid dates, a save failure and duplicate event delivery. A remote workflow call is not rolled back by the local database. Review RAP-04, RAP-05, BTP-07 and RICEFW-11.

Exercise 5 Read an EML response

A validation returns an error message in REPORTED but does not fill FAILED. A determination tries to fill FAILED. A create handler returns only the database key and loses the caller's content ID.

Your answer before reading the solution

Model answer

The validation needs to identify the rejected instance in FAILED and return the contextual message in REPORTED. Determinations do not expose FAILED; design expected rejection in an appropriate validation, precheck or action. Creation mapping must preserve the required correlation, including %cid where the contract requires it. Return the correct transactional identity in instance-related responses, including draft or preliminary context where applicable.

Follow-up

Explain %cid, %cid_ref, %tky and %pid without calling %tky a temporary key. For late numbering discuss preliminary-to-final mapping in ADJUST_NUMBERS. Review RAP-07 and RAP-10.

Exercise 6 Secure an approval action

An app hides the Approve button for ordinary users. Its DCL filters one base CDS entity, but the service uses a projection without its own access control. An ordinary user can call the action URL directly.

Your answer before reading the solution

Model answer

Implement the appropriate operation authorization for approval and keep business-state availability in feature control. Establish read authorization on the consumed projection, using supported DCL inheritance where appropriate. Check service access, instance scope and downstream APIs separately. Test direct requests and known keys as well as the UI. A value help or hidden field does not protect the data.

Follow-up

Would authorization: none or IN LOCAL MODE fix the bug? No: those deliberately bypass relevant checks in their defined contexts and require justification. Review CDS-07, ADV-02 and TEST-06.

Exercise 7 Classify Clean Core dependencies

Classify these hypothetical cases. Assume the stated release or nomination is verified for the target system; do not infer it from a name.

Case	Model classification and reasoning
ABAP Cloud extension using the appropriate released SAP APIs and released extension points	A for those dependencies and technologies
Classic code calling an API SAP has nominated as classic	B for that dependency
Read access to an internal SAP object that is neither released nor nominated and is not explicitly prohibited	C with upgrade risk and governance
Unsupported direct write to an SAP business table	D risk; replace with the supported business operation
Released custom wrapper calling a nominated classic API	The consumer sees a released custom contract; the underlying implementation still carries B dependency
BTP app using an unsupported private endpoint	Placement does not establish compliance; classify the actual dependency and remedy it

Follow-up

State the ATC mapping: A no corresponding usage finding, B priority 3, C priority 2, D priority 1. Explain why an ordinary security error in ATC is a different dimension. Review CORE-02 through CORE-16.

Exercise 8 Diagnose a failed batch and stale update

An OData batch returns HTTP 200 but one inner response is 412. The UI reports success and the developer retries with If-Match set to a wildcard.

Your answer before reading the solution

Model answer

Inspect all inner responses and identify the atomic group. HTTP 412 indicates a failed precondition, commonly a stale ETag. Reload current state and resolve the conflict instead of automatically overwriting a newer version. Verify whether sibling operations rolled back under the supported changeset or atomicity-group semantics. A wildcard is not a sound default repair for lost-update protection.

Follow-up

Explain why one batch can contain separate transaction scopes and why a CSRF token does not replace an ETag. Review ODATA-06 through ODATA-09.

Exercise 9 Build a reliable approval flow

A committed request event is delivered twice. Approval later arrives after the requester has withdrawn the document. The callback times out after the backend accepted it.

Your answer before reading the solution

Model answer

Use correlation and idempotency to avoid duplicate workflow creation. Validate current status, request version and actor on every decision. Reject a stale approval of a withdrawn request. A repeated callback for an already accepted decision should return a stable outcome; a timeout means the result may be uncertain, so reconcile before retrying an irreversible operation. Monitor failed deliveries and provide controlled recovery.

Follow-up

Draw the boundary between the local RAP transaction, event delivery, external workflow and callback transaction. Review RICEFW-11 through RICEFW-14 and BTP-08.

Exercise 10 Explain and test a refactoring

A report method selects data, calculates discounts, writes a file and commits the transaction. It uses a singleton for the current date and a nested SELECT in every item loop. What would you do first?

Your answer before reading the solution

Model answer

Establish existing behavior, callers and transaction ownership. Separate the pure calculation from data and I/O dependencies and inject a clock where it improves deterministic tests. Measure database execution count before redesigning the read path. Keep commit with the appropriate transaction owner and preserve required output semantics. Add focused regression cases and change incrementally rather than rewriting every statement.

Follow-up

Test discount boundaries, missing data, file failure and rollback behavior. Report which tests were actually run and which need the target system. Review OO-07, OO-13, TEST-01 and TEST-10.

Exercise 11 Recover from an uncertain HTTP result

Your ABAP caller sends a create-order request with JSON. The service commits the order, but the connection times out before the response arrives. The retry job proposes sending the same request again with no correlation identifier.

Your answer before reading the solution

Model answer

Treat the outcome as unknown, not as a proven failure. Determine whether the contract supports an idempotency key or a query by stable business correlation. Reconcile the result before retrying a non-idempotent creation. Validate Content-Type, response status and error payload independently. Preserve identifiers such as leading-zero customer numbers and distinguish missing fields, null and empty values.

Follow-up

What if the service returns 202? Explain how completion is observed. Explain why converting the JSON body to XML is a contract change, not just a header change. Review HTTP-01 through HTTP-12 and BTP-07.

Exercise 12 Make a parallel background job restartable

A classic background job sends three independent work partitions through aRFC. Worker A commits, B fails and C is still running when the coordinator times out. An operator wants to rerun the whole job.

Your answer before reading the solution

Model answer

Identify each partition and its actual business outcome. Do not assume that rollback in the coordinator undoes worker A. Collect callbacks and RFC exceptions, persist checkpoints under an explicit transaction design, and handle ambiguous outcomes through reconciliation. Bound parallelism by available resources and avoid overlapping locked records. Retry only safe unfinished units. Verify the effect of WAIT and task processing on the database LUW for the chosen implementation.

Follow-up

Distinguish the job scheduler from parallel execution. Explain the roles of SM36/SM37 in a classic landscape and application-job templates in supported cloud scenarios. State why job status and business item success are separate evidence. Review RFC-05 through RFC-12.

Exercise 13 Diagnose an IDoc or AIF error before resubmission

An inbound integration reports a mapping or application error. The business user sees the message in a configured AIF monitor. A similar earlier message already created a business document. An operator proposes repeated resubmission until the status turns green.

Your answer before reading the solution

Model answer

Correlate the message, external business key and existing document. Inspect the original IDoc control/data/status records and the configured AIF processing/error context. Correct the root cause with the proper authorization, then use the owning reprocessing mechanism. Preserve audit evidence and protect against a duplicate business effect or a stale-state overwrite. AIF is not a new transport that replaces all IDoc or OData behavior.

Follow-up

Explain how the diagnosis changes for transport hand-off versus application posting failure. Distinguish OData service semantics, IDoc asynchronous message processing and AIF monitoring/error handling. Review INTF-01 through INTF-12.

A 60 minute mock interview

Time	Interviewer prompt	What a strong response includes
0 to 5 minutes	Introduce yourself and one real project	Your actual role, constraint and outcome
5 to 12 minutes	Choose table types and explain a missing lookup	Key semantics, mutation and error handling
12 to 20 minutes	Diagnose a slow report	Evidence, SQL volume, access pattern and correctness
20 to 32 minutes	Design a request BO with draft and approval	Composition, transaction owner, messages and concurrency
32 to 40 minutes	Explain Clean Core levels and a missing API	Target edition, classification, options and governance
40 to 48 minutes	Discuss an OData or workflow failure	Protocol, retries, atomicity and security
48 to 55 minutes	Explain testing and deployment	Real evidence, negative tests, dependencies and limits
55 to 60 minutes	Ask questions about the role	Landscape, ownership, quality process and expectations

Score the interview

Score eight technical responses from 0 to 3. A 0 is wrong or missing; 1 is a definition with important gaps; 2 is correct with an example; 3 adds a relevant trade-off, failure mode and validation approach. Treat the score as a self-assessment rather than a hiring prediction. Choose the three weakest topics and repeat different questions from them tomorrow.

Personal story worksheet

For each of five real examples, write: business problem; my responsibility; constraints; options considered; decision; testing evidence; result; what I learned. Use this for a delivered feature, a defect, a performance problem, an integration and a disagreement about design. Use only outcomes and measurements that actually occurred.

Extension placement reference

Choose an extension approach

Start with the business requirement, supported contracts and transaction boundary

First	**Can standard functionality or configuration meet the need?**	Use the standard capability and avoid unnecessary custom code.
Key user	**Does a delivered business context cover the adaptation?**	Use supported custom fields, logic or UI adaptation tools.
On-stack	**Does the logic need local transactional consistency?**	Use released local contracts and suitable ABAP Cloud extensibility.
Side-by-side	**Does the process need an independent or cross-system app?**	Use supported remote APIs and events; design retries and consistency.

Placement does not determine the Clean Core level. Check dependencies separately.

Study plan

The main route is ABAP Cloud, RAP and Clean Core: 154 questions, including 96 senior scenarios. Allocate about 70% of preparation time to this track and 30% to supporting ABAP and integration. Use the topic library to understand each answer with its pictures, then switch to the separate practice mode to recall it without reading ahead.

Start with a senior diagnostic

Try CLOUD-01, CLOUD-04, SRAP-03, SRAP-08, SRAP-12, MODEL-01, MODEL-04, MODEL-14, OPS-05, OPS-11, GOV-02 and GOV-10. For each, explain the constraint, decision, rejected alternative and verification evidence. Score 0 for missing, 1 for partial, 2 for correct with an example, and 3 for a defensible trade-off with evidence. The two follow-up probes test whether your explanation survives a changed constraint.

A correct definition alone is not a senior answer. Mark a question ready only after explaining its failure boundary and a meaningful test. Review weak questions after one day, three days and one week. The browser records self-assessment; Anki supplies scheduled repetition if desired.

Use the web studio

Open `START_HERE.html`. Learn starts with the complete topic catalog. Select a topic or Browse all questions. Topic icons, a breadcrumb and the sidebar show your current subject. Filter this view by text or ID narrows the current selection; topic/subtopic filters and the naturally sorted question index provide other ways to browse. All difficulty levels stay together without badges on question cards.

Show all answers and Questions only display a check and dark background on the selected option. Your display preference is remembered locally. Open or close individual answers as needed; a mixture of open and closed answers is labeled Custom view. Compact source links sit directly beneath each question. Diagrams have a captioned preview and numbered switcher; Enlarge diagram opens zoom, the original file and PNG download. Senior criteria and probes expand under Go deeper.

Use Search all topics and questions, or Ctrl+K / Cmd+K, to search all 329 questions, answers, examples and source titles. This search ignores scoped topic, difficulty and practice filters. A result opens the question with controls to return to the results or view it in its topic. Browser Back and Forward follow your navigation.

Practice uses one session builder and defaults to Interview practice across all questions and levels. Choose a subject, All questions / New questions / Retry mistakes, and a session of 5, 10, 20 or all available questions. More settings contains optional difficulty and search filters. New means no attempt in the selected format; retry uses the latest incorrect quiz attempt or spoken self-rating of 0 or 1.

Interview practice covers all 329 questions: explain the answer aloud, reveal the model answer, then give yourself a score of 0-3. Multiple-choice quiz covers the 103 questions with authored choices and explains all four options after you submit. For RAP, the coverage is 52 interview questions and 24 multiple-choice questions. The format cards make this distinction visible before you start. Quiz XP and spoken readiness are tracked separately; no microphone recording or automatic spoken grading is used.

During a session, global search and opening a question preserve the session in memory. Resume practice returns to the same fixed question list and current position. This works within the same page; reload does not preserve an unfinished session.

Skip for now leaves the question unattempted. Finish session shows results for answers submitted so far. Results separate mistakes or weak self-ratings from questions not attempted; quiz accuracy uses submitted answers only. Retry unsuccessful answers, practice the unattempted questions, or choose New session. Practice these questions transfers a library selection into Interview practice. Pictures and references appear with the relevant model answers and quiz feedback. Downloads & progress contains optional PDFs, Anki decks and progress import/export. Export progress before moving browsers. See the usability and content review for the revision and verification record.

A daily hour

1. Recall weak questions for 10 minutes. 2. Study one Cloud segment and its visuals for 20 minutes. 3. Work through a senior scenario without the model answer for 15 minutes. 4. Take a 10-minute self-check, naming the evidence you would collect. 5. Record the weakest IDs and one unresolved release-specific assumption for 5 minutes.

If the interview is in 48 hours

Session	Work	Evidence of readiness
Day 1 morning	Cloud architecture 16 and Clean Core 08/20	Defend extension placement, API contracts and a bounded exception
Day 1 afternoon	RAP 05/06/17	Trace buffer, save, numbering, draft, concurrency and failure recovery
Day 1 evening	CDS/security 04/18	Diagnose a DCL leak, unsupported filter and wrong count
Day 2 morning	Production 19 and weak foundations	Design meaningful tests and an uncertain-remote-outcome recovery
Day 2 afternoon	Senior mock: 75 minutes plus review	Score all six cases and repair the lowest dimensions
Final review	Quick revision and truthful project stories	Give a 90-second decision with an example and evidence

Seven day route

Day	Main work	Practice
1	16 Cloud architecture; 08 Clean Core baseline	Placement, runtime versus model, C0/C1/C2 versus A-D
2	05 RAP foundations; first half of 17	Transaction/save ownership, bulk EML and field presence
3	06 advanced RAP; remaining 17	Draft resume, entity/total ETags, cross-BO constraints
4	04 CDS; 18 security and queries	Proposed-state checks, DCL entry point, count/paging/filter
5	19 operations; 20 governance	Real BO tests, incidents, compatibility and exception retirement
6	Foundations 01/21, OO 02/22, patterns 23; integration by diagnostic need	HTTP/REST/JSON/XML, OData, RFC/BAPI/LUWs/jobs, IDoc/AIF
7	Six-case senior mock; 12 real project stories	Repair weak answers and explain two personal trade-offs

Fourteen day route

Days	Focus	Output
1-2	Cloud architecture and Clean Core levels	Capability matrix and extension decision
3-5	RAP foundations and senior transactions	Save/concurrency drawings and three failure tests
6-7	CDS, authorization and custom queries	Security matrix and query-contract test cases
8-9	Production, testing and performance	Evidence plan and incident recovery walk-through
10	Clean Core governance	Bounded exception and upgrade/retirement plan
11-12	Foundations 21, OO 22, patterns 23 and delivery/HANA 24	Trace value/reference behavior, defend a pattern and verify integration recovery
13	Senior mock without notes	Six scored cases and weakest three competencies
14	Weak cards, quick revision and project stories	Two-minute truthful examples with measured outcomes

Revisit earlier mistakes on days 4, 8 and 12. Read baseline modules when the senior scenario exposes a gap; do not replace missing foundations with memorized architecture phrases.

Rehearse truthful project stories

Prepare a two-minute story for a delivered feature, a production defect, a performance improvement, a difficult integration and a design trade-off. For each, record the business context, your role, constraint, decision, verification and outcome. Use your actual projects and measurements. Clearly distinguish a learning exercise from production delivery.

Import the flashcards

Import `flashcards/ABAP_Interview_Anki.tsv` into Anki as UTF-8, tab-separated text. The file has no header row. Map column 1 to Front, column 2 to Back and column 3 to Tags; enable HTML in fields. Use a Basic note type. Question IDs are included in the front so updates can be matched. The HTML Study Desk also works without Anki: download the repository and open `START_HERE.html` locally.

Release context to clarify

Before discussing a concrete implementation, establish SAP edition, release and support package, ABAP language version, API release status, consumer and SAPUI5 version. Classic transactions in the guide are examples for landscapes where those tools are available. The date of this study pack is not a claim that every feature exists on every target system.

Learn with visual explanations

In START_HERE.html, open a question's answer to see its linked visual previews. Select a numbered diagram and use Enlarge diagram to inspect it without leaving the question. Interview practice reveals diagrams with the model answer; Multiple-choice quiz makes them available under Full explanation & diagrams after checking. The GitHub gallery and printable atlas remain available for browsing the full collection; the app's Downloads & progress menu links to both.

For each picture, first describe what is happening without reading the caption. Answer its recall prompt, then state where the analogy or model stops being accurate. Redraw one example from memory: a join result, a stale ETag update, the RAP artifact stack or the Clean Core levels. Reopen the related questions to check the explanation.

Illustrated Anki deck

Use `flashcards/ABAP_Interview_Illustrated_Anki.tsv` for the same complete question set with images. Copy the PNG files from `flashcards/media/` directly into your Anki profile's `collection.media` directory, preserving filenames. Import with Front / Back / Tags mapping and HTML enabled. Do not add a media subdirectory.

Choose one deck for the first import. Both use identical front fields, so use Anki's matching/update options when replacing an existing deck. Check the import preview and verify that `RAP-12` displays its images.

Reference: Anki text and media import manual.

Focused senior Anki deck

Use `flashcards/ABAP_Cloud_Senior_Anki.tsv` for only the 96 senior scenarios. It includes follow-up probes, strong-answer criteria and verification evidence. It is a subset of both full decks and uses the same front fields; choose one initial import or use Anki matching/update options.

Focused foundations and pattern route

Start with BASE-01/02/04/06/12/13/16, OODEEP-01/03/06/09/12 and PAT-01/03/04/06/08/09/12. Predict the worked example before reading it. For each pattern, name the variation or failure boundary it solves and the simpler alternative you rejected. Complete a challenge, explain all three distractors, then repeat the same decision aloud without options. Read the quiz guide and current market coverage to choose remaining gaps.